

Integrating Molecular Graph Data of Drugs and Multiple -Omic Data of Cell Lines for Drug Response Prediction

Giang T.T. Nguyen[✉], Hoa D. Vu[✉], and Duc-Hau Le[✉]

Abstract—Previous studies have either learned drug's features from their string or numeric representations, which are not natural forms of drugs, or only used genomic data of cell lines for the drug response prediction problem. Here, we proposed a deep learning model, GraOmicDRP, to learn drug's features from their graph representation and integrate multiple -omic data of cell lines. In GraOmicDRP, drugs are represented as graphs of bindings among atoms; meanwhile, cell lines are depicted by not only genomic but also transcriptomic and epigenomic data. Graph convolutional and convolutional neural networks were used to learn the representation of drugs and cell lines, respectively. A combination of the two representations was then used to be representative of each pair of drug-cell line. Finally, the response value of each pair was predicted by a fully connected network. Experimental results indicate that transcriptomic data shows the best among single -omic data; meanwhile, the combinations of transcriptomic and other -omic data achieved the best performance overall in terms of both Root Mean Square Error and Pearson correlation coefficient. In addition, we also show that GraOmicDRP outperforms some state-of-the-art methods, including ones integrating -omic data with drug information such as GraphDRP, and ones using -omic data without drug information such as DeepDR and MOLI.

Index Terms—Drug response prediction, integrating -omic data, graph convolutional network, convolutional neural network

1 INTRODUCTION

DUe to significant developments in high-throughput screening technologies, personalized medicine has gained much attention in discovering the right therapy for the individual patients. This procedure needs a systematic study of drug responses through a large number of patient profiles [1]. However, because of time-consuming, financial, and practical reasons in the scope of testing drugs on patients, there are a few projects providing drug response data on patients [2]. Therefore, predicting drug response on the patient remains challenging. To address this issue, cell lines derived from clinical tumor samples have been used as artificial patients in various existing computational studies to estimate the response on patients.

In recent years, high-throughput projects such as CCLE [3], GDSC [4], NCI [5] have provided different types of -omics data of a panel of cell lines (including gene expression, somatic mutation and copy number aberration, methylation, and so on) as well as their response against hundreds

of drugs. These high throughput databases have facilitated the development of computational methods for predicting drug response for cell lines. A range of machine learning-based approaches for drug response prediction has already been proposed, including traditional models such as linear regression, support vector machines (SVMs), and neural network models [6], [7], [8], [9], [10], [11]. In addition, more advanced models like matrix factorization, multitask-learning [12], [13], [14], [15], [16], ensemble learning strategies [17], [18], as well as collaborative filtering techniques used in the recommender systems [19], [20] have been also proposed. Drugs and cell lines are often represented in high-dimensional forms, e.g., -omics data of tens of thousands of genes were generated for each cell line, and a large number of chemical and structural features were calculated for each drug. Meanwhile, the sample size is small since nearly thousands of cell lines were tested against hundreds of drugs/compounds [21], [22], [23]. Thus, machine learning methods often face with the "small n, large p" challenge and lead to the limitation of their prediction performance [23], [24]. In addition, there were network-based ones that were relying on similarity networks (structural similarity between drugs and biological similarity between cell lines), and known drug-cell line responses have been proposed [25], [26].

Recently, deep learning as a subclass of machine learning approaches has increasingly been a popular solution in drug discovery studies [27]. With the ability to integrate high and low dimensional representations of input data such as -omic data and other complex biological data, as well as handling noisy data, the deep learning-based methods can make more accurate predictions than other machine learning ones [28], [29], [30]. Especially, in the drug

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response problem, deep learning is utilized to automatically learn genomic features of cell lines and the structural features of drugs to predict anti-cancer drug responsiveness, thus, reduces the need for calculating molecular features or feature selection before training. A diversity of neural network architectures has been applied to predict drug sensitivity [31], [32], [33], [34], [35], [36], [37]. Some of them [31], [32] used AEs, VAEs techniques included encode and decode networks to learn -omic features as reduced dimension representations to estimate drug response. Also, using -omic data without drug representations to predict responses, DeepDR [37] and MOLI [34] encoded omic features (including somatic mutation, copy number aberration, and mRNA gene expression data), into subnetworks and the learned features are concatenated into one representation to predict drug responses. However, unlike DeepDR [37], MOLI [34] is a drug-specific model. Other improved deep learning methods are based on Convolutional neural network (CNN) techniques that can predict drug response by integrating some omic data of cell line and drug information. These drug features are represented as either compound images or string or numerical forms [28], [35], [38], [39], [40], [41]. In tCNNS [28], drug features were extracted from SMILES string representations. Moreover, genomic features and these string representations of drugs were represented as one-hot encoded representations in each subnetwork, then were linked to a final prediction network. In summary, previous studies have neither used drug information nor used them in a string or numerical representations that are not a natural molecular graph form of drugs; thus, the structural information of drugs may be lost.

Graph convolutional networks (GCN) are able to learn representations of compound structures depicted as molecular graphs and have been used in various other drug discovery prediction tasks. However, there are few studies that applied this model in drug response prediction (DRP). GraphDTA [41] had been proposed to predict drug-target affinity. This is referred to as the first model learned drug features from graph representations of drugs by using a graph convolutional network. The model achieves the best performance compared to other deep learning-based methods that represented drugs as strings in the task of drug-target binding affinity prediction. Similarly, In our previous study, GraphDRP [42], drug features were learned from their molecular graphs by capturing the bonds among atoms. However, it only used genomic data (i.e., mutation and copy number aberration) to predict drug response but did not integrate other -omic data such as transcriptomic (i.e., gene expression) or epigenomic (i.e., methylation) of cell lines. Besides, previous studies indicated that transcriptomic data were the most informative dataset in many approaches [22], [43], [44] for drug response prediction. When combined with transcriptomic data, the other -omic data can enhance prediction performance [22].

In this study, in addition to taking benefit of graph convolutional network to learn drug's features from their molecular graphs as in GraphDRP [42], we proposed a method, namely GraOmicDRP, to integrate multiple -omic data to predict drug response for cell lines. In GraOmicDRP, drug compounds obtained from the SMILES string format were represented as graphs of the bonds among atoms. These

graph representations were then learned and converted to 128-dimension vectors via GCN block. Especially, by integrating multiple -omic profiles of cell lines in different CNN blocks, we could extract cell line representations of not only genomic data (i.e., mutation and copy number aberration (MUT_CNA)) as in GraphDRP [42], but also other kinds of -omic data of cell lines (i.e., gene expression (GE), methylation (METH)). Moreover, cell line features are represented as vectors that are learned then also are converted to 128-dimension vectors in individual CNN blocks. Finally, these drug and cell line representations are concatenated altogether before putting into two fully connected layers to evaluate drug response values. The experiments were performed by single -omic and integrating multiple -omic data. For mixed test (i.e., drug-cell line pairs were randomly split, but drugs and cell lines in the testing phase may be in the training phase), experimental results show that gene expression (GE) outperformed other single -omic models including genomic data (i.e., mutation and copy number aberration, MUT_CNA) that was used in GraphDRP [42]; meanwhile, the combinations of GE & METH and GE & MUT_CNA data are the best overall in terms of root mean square error (RMSE) and Pearson correlation coefficient (CCp), respectively. For bind-cell line test (i.e., drug-cell line pairs were randomly split, and cell lines in the test phase did not appear in the training phase) and blind-drug test (i.e., drug-cell line pairs were randomly split, and drugs in the test phase did not appear in the training phase), GraOmicDRP was also superior to GraphDRP [42] in terms of both RMSE and CCp. In comparison with other state-of-the-art methods, which did not use drug information, we also showed that our method was better than DeepDR [37], where only GE was used. When acted as a classifier, GraOmicDRP also outperformed the drug-specific model, MOLI [31], overall four selected drugs (i.e., Docetaxel, Erlotinib, Gemcitabine, and Paclitaxel) in terms of AUC values.

2 MATERIALS AND METHODS

2.1 Materials

Large-scale screening cell line projects such as CCLE [3], GDSC [4], have generated not only -omic data of cell lines but also drug response data for anti-cancer drugs on thousands of cell lines. The two databases contain gene expression (i.e., transcriptomic data), which indicates the quantity of RNAs transcribed from DNA and thus the number of translated proteins in a cell. Therefore, the expression level of a gene indicates the activity level of a gene in a certain state (e.g., diseased, or normal) in a cell. In addition, genomic data such as mutations and copy number aberration in the genome and epigenomic data were also generated. Besides, cell lines are cultured and treated with different doses of drugs. The drug response measure of a particular drug normally is referred to as IC50 value, which indicates the dose of a particular drug needed to inhibit the biological activity by half. These continuous response values are used in regressors to predict drug responses for cell lines. For classifiers, these values are discretized into binary form (i.e., sensitive and resistant) by applying the waterfall method [3].

In fact, GDSC is the largest database, including a diversity of -omic data and drug sensitivity information for

TABLE 1
Datasets

	Datasets	# Cell lines	# known drug-cell line responses
Single -omic	METH	676	150,761
	GE	857	191,034
	MUT_CNA	857	191,049
Multiple -omic	GE & METH	663	147,891
	GE & MUT_CNA	838	186,864
	METH & MUT_CNA	676	150,761
	ALL (GE & METH & MUT_CNA)	663	147,891

thousands of cell lines and hundreds of drugs, which are much larger than the number of tested cell lines and tested drugs in CCLE. Therefore, we selected the GDSC dataset for our study. More specifically, as in previous studies [34] and GraphDRP [42], we used the GDSC dataset with 223 drugs, 948 cell lines, and drug response values in terms of IC50 normalized in a range (0,1) as well as drugs' SMILES format [42], [36]. Additionally, we used three types of -omic data, including gene expression (GE), mutation and copy number aberration (MUT_CNA), and methylation (METH), of cell lines. MUT_CNA data are in a binary form representing whether or not a gene contains a mutation or copy number aberration. Similarly, methylation data are also binarized to represent whether a gene is hyper-methylated or hypomethylated. Meanwhile, GE provides continuous expression level for genes; thus, GE values are normalized in a range (0,1). Table 1 shows the number of cell lines having single -omic data or combinations of them (i.e., multiple -omic data) and corresponding the number of known drug-cell line responses.

2.2 GraOmicDRP

In this study, we proposed a model, namely GraOmicDRP, that combines chemical structural data of drugs and a single or multiple -omic data of cell lines to predict response values. Fig. 1 illustrates the proposed model.

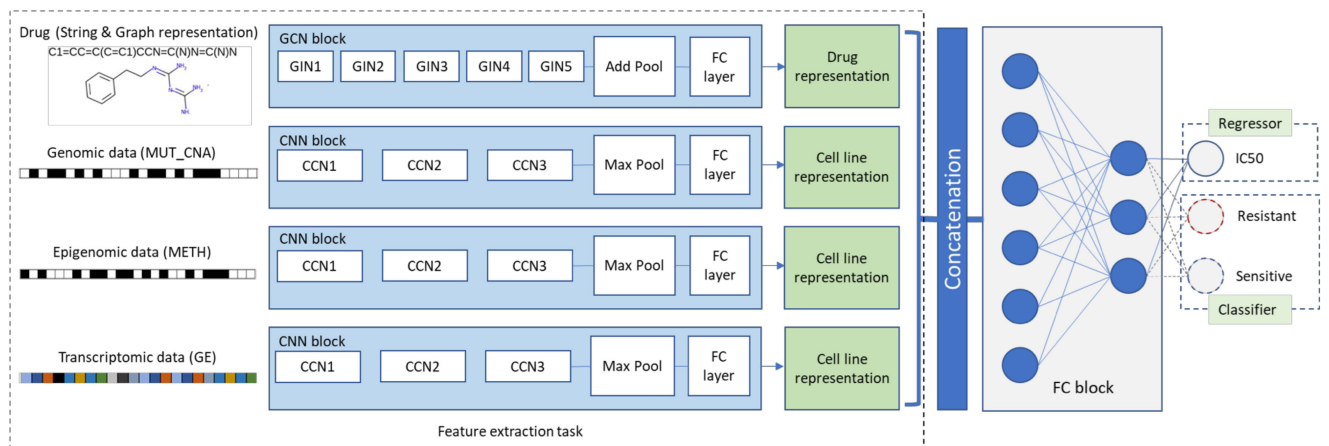


Fig. 1. GraOmicDRP integrate drug and multi-omic data for predicting drug response for cell lines. A GCN block is used to learn features of drugs from their graph representation, meanwhile CNN blocks are applied to extract features from individual omics data for cell lines. Outputs of these blocks are flattened to be 128-dimension vectors of drugs and cell lines. Then, these 128-dimension vectors are concatenated and put through two FC layers to predict the response values. When the response values were continuous, GraOmicDRP was acted as a regressor, meanwhile it can be played as a classifier for predicting binary form of the response values (i.e., sensitive, and resistant).

Similar to our previous method, GraphDRP [42], firstly, drugs represented as SMILES string format [45] were converted to molecular graphs reflecting interactions between the atoms inside the drugs by using RDKit [46]. To learn the drug's features, we used a GCN block to extract drug representations. Currently, there exist several powerful neural network architectures on graphs such as GCN [47], Graph Attention Network (GAT) [48], Graph Isomorphism Network (GIN) [49], and combined GAT-GCN applied for learning drug's features from their graph representation [41]. Nevertheless, GIN was evaluated as the best one in both drug-target affinity prediction, GraphDTA [41], and drug response prediction, GraphDRP [41]. Therefore, we adopt GIN as the main component in GCN block to learn the drug's features in our model.

In GIN, the node feature is updated by multi-layer perceptron (MLP) as

$$MLP = \left((1 + \mu) x_i + \sum_{j \in N(i)} x_j \right) \quad (1)$$

where μ is either a learnable parameter or a fixed scalar value, x is the node feature vector, and $N(i)$ is the set of nodes neighboring to i .

In this study, we deployed five GIN layers in the GCN block. In which, MLP includes two linear layers instead of one linear layer as a suggestion of [49] to make the model distinguish drug graphs more effectively. ReLU activation function and batch normalization layer were used after each layer. Then, a global add-pooling layer was added to combine a graph representation vector, and finally, a fully connected layer flattened the results to 128 dimensions (Fig. 1).

For learning cell line features, instead of using only genomic features of cell lines (i.e., mutation and copy number aberration) as in GraphDRP [42], GraOmicDRP can integrate multiple -omic data types (transcriptomic and epigenomic) to extract more cell line representations for the prediction. A convolutional neural network is a technique that can detect relevant patterns in data [47]. Since the -omic

features of cell lines are represented as 1D feature vectors for cell lines, these vectors can be used as input into a 1D convolutional neural network (CNN) to learn latent features on those data. Layers of 1D convolutions and pooling are used to extract hidden patterns in data. Therefore, for each type of the -omic data, we used an individual CNN block to learn cell line features. More specifically, a CNN block consists of three convolutional layers with a max-pooling layer and Rectified Linear Unit (ReLU) as the activation function. Then the output was flattened to a 128-dimension vector of cell line representation. The model is flexible for using single -omic data as well as integrating multiple -omic data.

After that, we concatenated drug representation and cell line representations to two fully connected (FC) layers to predict continuous drug response values (i.e., IC50 values). For this case, GraOmicDRP was played as a regressor. The number of nodes of the first FC is 1024, and those of the second FC is 128, the input for this FC block is a vector of 256 or 384 or 512 dimensions depending on the setting for single -omics, pair of -omics, or all -omics integration, respectively. Furthermore, for predicting whether a cell line is sensitive or resistant against a drug, we reconstructed and retrained the model to be a classifier with the binary form of the responses.

2.4 Performance Evaluation

Testing scenarios

We evaluated the prediction performance of GraOmicDRP and the competitive ones in three scenarios. First, a mixed test was performed by shuffling data randomly, then splitting the data in known pairs of drug responses into training, validation, and testing sets, respectively in order to keep the model from general as well as reducing overfitting. In short, the mixed test, drugs, and cell lines in the training phase may appear in the testing phase. The known pairs are split into 80 percent as the training set, 10 percent as the validation set, and 10 percent as the testing set. While the validation set is used to modify the hyperparameters of the model in the training phase, the testing set is used to evaluate the performance of the model. In the mixed test, the prediction performance of GraOmicDRP was assessed for single -omic data, including MUT_CNA used in GraphDRP. In addition, the performance of different combinations of -omic data were also evaluated. The best combination was then used to evaluate the performance of GraOmicDRP and to compare it with GraphDRP on other scenarios. More specifically, the second scenario was performed by a blind-cell line test, i.e., when it is required to make predictions for a new cell-line that is not in the training phase. So, cell lines in the training phase did not appear in the testing phase. A total of 90 percent cell-lines are randomly selected, and their response values are kept for the training phase, including 80 percent cell lines for the training set and 10 percent cell lines for the validation set. The remaining 10 percent cell lines is used as the testing set. The third scenario was that when we need to predict responses for unseen drugs, thus, we performed a blind-drug test. Similarly, a total of 90 percent of drugs and their response values are randomly selected for training, including 80 percent drugs for the training set and 10 percent drugs for the validation set. The remaining set, 10 percent of drugs, are used as the testing set.

Performance measurements

GraOmicDRP as a regressor: The prediction performance of each method was assessed by comparing the predicted values (Y) to the observed (known) values (O) using Root Mean Squared Error (RMSE) and Pearson correlation coefficient (CC_p) metrics.

Root Mean Squared Error (RMSE) measures the difference between the predicted values to the observed values as formulated:

$$RMSE = \sqrt{\frac{1}{n} \sum_1^n (o_i - y_i)^2} \quad (2)$$

where o_i and y_i are as observed response and predicted value of the i^{th} sample, respectively. n is the number of samples.

Pearson correlation coefficient measures how strong a relationship is between observed values (O) and predicted values (Y). With σ_o and σ_y are the standard deviation of the observed and predicted values, respectively, CC_p is defined as follows:

$$CC_p = \frac{\sum_1^n (o_i - y_i)^2}{\sigma_o \sigma_y} \quad (3)$$

GraOmicDRP as a classifier: The prediction performance of a binary classifier is assessed by an AUC value (i.e., the area under the receiver operating characteristic (ROC) curve). The ROC curve plots the relationship between True Positive Rate (TPR) and False Positive Rate (FPR) at all classification thresholds [46].

$$TPR = \frac{TP}{TP + FN} \quad (4)$$

$$FPR = \frac{FP}{TP + FP} \quad (5)$$

where TP and FP are respectively a number of true and false positives (i.e., sensitive responses); meanwhile TN and FN are the number of true and false negatives (i.e., resistant responses).

3 RESULTS

In this study, we proposed a method, GraOmicDRP, to integrate molecular graph data of drugs and multiple -omic data of cell lines for drug response prediction. The method allows incorporate up to three -omic data types (i.e., genomic (MUT_CNA), epigenomic (METH), and transcriptomic (GE)) with drug information.

Integrating -omic data with drug information

In the previous study, we showed that GraphDRP outperformed tCNNS [41], in which drug features were learned from SMILES string representations by convolutional neural networks, using MUT_CNA data. In this section, the prediction performance of GraOmicDRP was assessed and compared with GraphDRP, based on different testing scenarios (See section Performance Evaluation).

Mixed test: For the mixed test, we firstly assessed the prediction performance on single -omic data (i.e., METH, GE, and MUT_CNA), in which GraphDRP was performed on MUT_CNA. Experimental results show that GE achieved

TABLE 2
Performance Comparison in the Mixed Test Assessed On
Known Pairs of Drug-cell line Responses

Methods		EVALUATION	
		RMSE	CCp
Single -omic data	METH	0.0279	0.9104
	GE	0.0259	0.9165
	MUT_CNA	0.0263	0.9120
Multiple -omic data (GraOmicDRP)	GE & METH	0.0239	0.9310
	GE & MUT_CNA	0.0246	0.9236
	METH & MUT_CNA	0.0252	0.9277
	ALL	0.0244	0.9295

The best performances are in bold. (GraphDRP) uses single -omic data MUT_CNA).

TABLE 3
Performance Comparison in the Mixed Test Assessed on Indi-
vidual Drug's Performance

Methods		EVALUATION	
		RMSE	CCp
Single - omic data	METH	0.0278	0.5770
	GE	(± 0.0087)	(± 0.1483)
	MUT_CNA	0.0254	0.6603
		(± 0.0073)	(± 0.1531)
Multiple - omic data	MUT_CNA	0.0244	0.6512
		(± 0.0088)	(± 0.1595)
	GE & METH	0.0225	0.7084
		(± 0.0070)	(± 0.1584)
	GE & MUT_CNA	0.0229	0.6900
	METH & MUT_CNA	(± 0.0072)	(± 0.1534)
	METH & MUT_CNA	0.0236	0.6858
		(± 0.0077)	(± 0.1542)
	ALL	0.0234	0.6835
		(± 0.0077)	(± 0.1605)

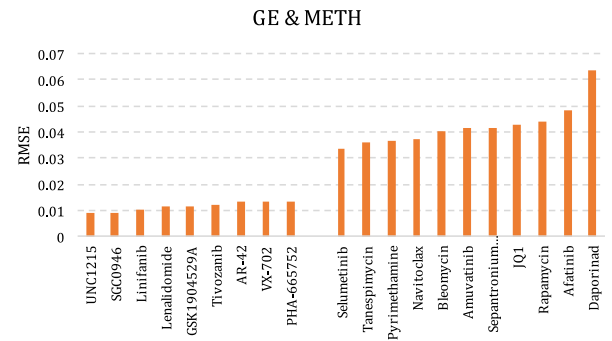
The best performances are in bold.

the best performance in terms of both RMSE (0.0259) and CCp (0.9195) (Table 2); meanwhile, those for GraphDRP (i.e., MUT_CNA) are 0.0263 and 0.9120, respectively. This result indicates that gene expression (GE) is the most informative compared to other -omic data when they are used for drug response prediction. This is because gene expression is a type of molecular phenotype which is closer to how cell lines respond to drugs.

Next, we further integrated multiple -omic data to find which combination performs best. More specifically, we integrated pairs of -omic data (i.e., GE & METH, GE & MUT_CNA, and METH & MUT_CNA) and all together. Table 2 shows that the combination of transcriptomic and epigenomic data (GE & METH) is the best in terms of RMSE (0.0239) and CCp (0.9310). All the combinations are also better than any single -omic cases. Interestingly, the combination without GE (METH & MUT_CNA) achieved the worse prediction performance among multiple-omic combinations in terms of RMSE. This once again indicates that gene expression data were the richer information in representing the biological characteristics of cell lines in drug response prediction.

Different drugs may have different prediction performance, thus we calculated RMSE and CCp for each drug.

A



B

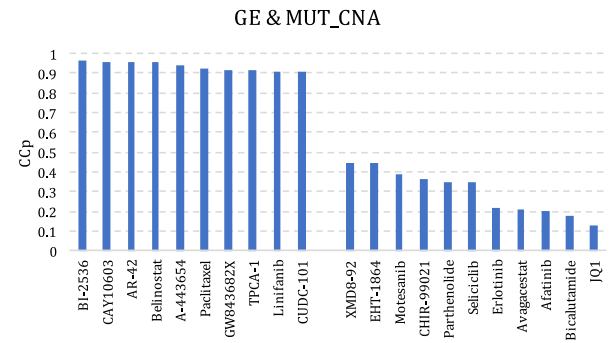


Fig. 2. Ten drugs obtained the highest (on the left) and lowest (on the right) performance. (A) Combination of GE & METH achieved best performance in terms of RMSE. (B) Combination of GE & MUT_CNA achieved best performance in terms of CCp.

Then, the overall prediction performance of GraOmicDRP was calculated as an averaging value of all drugs (Table 3).

Table 3 shows that the overall prediction performance of GraOmicDRP is consistent with that in Table 2 for assessing overall known drug-cell line pairs. Indeed, GE is the best among three single -omic data; meanwhile, the combinations of GE & METH archived the highest performance followed by GE & MUT_CNA in terms of RMSE and CCp, for multiple-omic data. Fig. 2 shows ten drugs that achieved the highest and lowest performance when combining GE & METH (Fig. 2A) and GE & MUT_CNA (Fig. 2B) in terms of RMSE and CCp, respectively.

Blind-cell line test: The goal of the blind-cell line test is to predict response value for unseen cell lines (i.e., cell lines that were not in the training phase). To compare with the best result of GraphDRP (i.e., GraphDRP-GIN, GIN model was used to learn drug's features), we chose the best combination of -omic data in terms of RMSE (i.e., GE & METH) for GraOmicDRP. A total of 90 percent cell lines are randomly selected, and their drug response values are used for the training phase. The remaining cell lines are kept for the testing phase. The performance comparison in terms of both CCp and RMSE for the blind-cell line test indicates that our method is better than GraphDRP. Table 4 shows that GraOmicDRP archived smaller RMSE (0.0327) and higher CCp (0.8766) than those of GraphDRP.

Blind-drug test: The goal of the blind-drug test is to predict response values for unseen drugs (i.e., drugs that were not in the training phase). Similar to the blind-cell line test, we used the best -omic data combination for GraOmicDRP.

TABLE 4
Performance Comparison in the Blind-Cell Line Test Assessed on Known Pairs of Drug-Cell Line Responses

Methods	EVALUATION	
	RMSE	CCp
GraphDRP-GIN	0.0363	0.8460
GraOmicDRP (GE & METH)	0.0327	0.8766

The best performances are in bold.

TABLE 5
Performance Comparison in the Blind-Drug Test Assessed on known Pairs of Drug-Cell line Responses

Methods	EVALUATION	
	RMSE	CCp
GraphDRP-GCN	0.0542	0.3241
GraOmicDRP (GE & METH)	0.0590	0.4309

The best performances are in bold.

(GE & METH) to compare with the best case of GraphDRP (i.e., GraphDRP-GCN, GCN model was used to learn drug's features). More specifically, 90 percent drugs are randomly selected, and their response values are used for the training phase. The remaining drugs are kept for the testing phase.

Table 5 shows that our method achieved comparable performance in terms of RMSE (i.e., 0.0542 and 0.0590 for GraphDRP-GCN and GraOmicDRP, respectively), but better in terms of CCp (i.e., 0.3241 and 0.4309 for GraphDRP-GCN and GraOmicDRP, respectively).

Taken together, experimental results show that GraOmicDRP is better than GraphDRP for all testing scenarios. This indicates the importance of integrating multiple -omic data, especially the transcriptomic (GE) data, for drug response prediction problems.

Integrating -omic data without drug information

Beside tCCNs [25] and GraphDRP [42], which used -omic data with drug information, we here compared the prediction performance of our method with other two state-of-the-art methods, DeepDR [37] and MOLI [34], which did not integrate drug information in their models. DeepDR [37] is a regression model integrating -omic data (including mutation and gene expression) without the chemical structure of drugs to predict drug responses. Consistently with our study, gene expression (GE) was found to be more directly to biological function than mutation and thus the prediction performance of DeepDR [37] on GE was comparable with that on the combination of GE and MUT. Thus, following architecture and procedure of DeepDR [37], we rebuilt and retrained DeepDR [37] based on gene expression dataset from TCGA [50] and then evaluated it on the gene expression dataset from GDSC [4] as in our blind-cell line test method. Experimental result shows that DeepDR [37] achieved 0.0330 of RMSE and 0.8722 of CCp; meanwhile, those for GraOmicDRP were 0.0327 and 0.8791, respectively (See the result of Single -omic data for GE in Table 2). The results indicate that our method outperformed DeepDR [37].

Besides the regression models predicting drug response as continuous response values (i.e., IC50), knowing how a

TABLE 6
Performance Comparison With Moli

		Docetaxel	Erlotinib	Gemcitabine	Paclitaxel
MOLI (GE & MUT_CNA)		0.6438	0.7295	0.5710	0.6500
Gra-OmicDRP	GE & METH	0.7000	0.8304	0.8643	0.8704
	GE & MUT_CNA	0.7304	0.8194	0.8159	0.9444
	MUT_CNA & METH	0.7281	0.6637	0.6773	0.7407
	ALL	0.7414	0.7708	0.8362	0.9259

The best performances for each drug are in bold.

cell line sensitive or resistant to a drug could be more meaningful. Thus, the continuous response values are binarized as sensitive and resistant, and classification models have been proposed. In addition, instead of building a general-purpose model for all drugs, constructing the drug-specific models is also useful. MOLI [31] is a drug-specific classification model that integrated multi-omic data (MUT, CNA and GE) of cell lines without chemical structure information of drugs. More specifically, MOLI [34] constructed separate models for six drugs (i.e., Docetaxel, Cisplatin, Gemcitabine, Paclitaxel, Erlotinib, and Cetuximab), then tested on PDX [51]/TCGA [2] datasets for those drugs.

To compare the prediction performance with MOLI [31], we redesigned GraOmicDRP as a classifier (See Section 2.2). For a fair comparison, we rebuilt the pre-trained MOLI [34] model and tested MOLI [34] and our method on the same testing set from GDSC. Among the six drugs obtained in MOLI [31], only four drugs (i.e., Docetaxel, Erlotinib, Gemcitabine, and Paclitaxel) are available in the GDSC database. Therefore, we compared the two methods based on these four drugs. Table 6 illustrates that GraOmicDRP outperforms MOLI [34] for all the drugs in terms of AUC values.

4 CONCLUSIONS AND DISCUSSION

In this work, we proposed a deep learning-based method to integrate molecular graph of drugs and multiple -omic data of cell lines for drug response prediction. In addition to taking benefit of learning drug's features by their graph representation with GCN model, we integrated multiple -omic data of cell lines to improve the prediction performance by CNN models. Indeed, experimental results show that by integrating drug graph representation with combining informative data such as transcriptomic data with other -omic data has improved the performance of drug response prediction in terms of both RMSE and CCp. For single -omic data, the transcriptomic data showed to be the best compared to the genomic and epigenomic data. For multiple -omic data, combinations of the transcriptomic and epigenomic data were superior to other combinations. In addition, we showed that our method outperformed GraphDRP in different testing scenarios illustrating real situations such as the mixed test for untested drug-cell pairs, the blind-cell line test for untested cell lines (e.g., new cell lines), and the blind-drug test for untested drugs (e.g., new

drugs). Moreover, to investigate the effectiveness of integrating graph representation of drugs with multiple -omic data of cell lines in predicting drug response, we compared the prediction performance between GraOmicDRP and other two state-of-the-art methods, DeepDR [37] and MOLI [34], which do not use drug information in their models. The results indicate that our method is superior to the two methods on the transcriptomic data and for the four selected drugs, respectively.

The results of the mixed test, the blind-cell line test, and the blind-drug test also show that the prediction performance for the mixed test is better than that of the blind-cell line test, and the prediction performance of the blind-cell line test is superior to that of the blind-drug test. The reason is that when comparing the mixed test with the other tests, drugs and cell lines in the testing phase may appear in the training phase for the mixed test; meanwhile, they were not in the other tests. Comparing the blind-cell line test and the blind-drug test, although drugs/cell lines in the testing phases are not in the training phases, cell lines in the testing phase in the blind-cell line test may belong to the same tissues with those in the training phase; thus, they may share similar biological characteristics. Therefore, the prediction performance for the blind-cell line test is generally better than that for the blind-drug test. Therefore, in future works, it will be interesting to assess the proposed method with blind-tissue or blind-cancer subtype tests. In other words, we should assess how well our method can predict drug responses for unseen tissues/cancer subtypes. Furthermore, the number of known drug-cell line pairs is large enough for randomly splitting them into training/testing sets for performance evaluation, future study should apply a k-fold cross-validation scheme to make the assessment more precise when assessing the prediction performance of the models, though. Finally, the prediction of drug combination/drug synergy [52], [53], [54] could be improved by learning drug's features from their graph representation and integrating multiple -omic data of cell lines.

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